Thummim Mekuria

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SUMMARY

Astrophysics PhD candidate at UC Santa Cruz skilled in developing physical models, parallel computing pipelines, and statistical modeling to analyze a diversity of datasets, including hyperspectral observations, 3D hydrodynamical simulations, and agricultural commodity prices. I can communicate with both expert and general audiences, and have designed curricula for teaching physics, programming, and data analysis to undergraduate students. Seeking a position as a Quantitative Analyst, Research Scientist, or Machine Learning Engineer.

EDUCATION

University of California, Santa Cruz, *PhD, Astronomy & Astrophysics* **Expected 2027**
M.Sc., Astronomy & Astrophysics **2025**
Pomona College, *B.A., Astrophysics & Philosophy* **2021**

- Awarded Distinction in Senior Exercise (2021) and Pomona College Scholar (2017, 2021)

SKILLS & CERTIFICATIONS

- Quantitative modeling: developing physical models of gas kinematics in the local and early universe; fitting models to data
- Programming: Python (incl. NumPy, Pandas, Astropy, SciPy), Jupyter, Fortran (intermediate), Git/GitHub, parallel computing, memory management for simultaneous analysis of multiple large (> 100 GB) datasets
- Science communication: poster and verbal presentations for technical and general audiences; curriculum design
- Languages: Amharic, English, French (fluent); Arabic (intermediate)
- Certifications: The Erdős Institute, Quantitative Methods in Finance (2026)

WORK EXPERIENCE

University of California, Santa Cruz: *Santa Cruz, CA* **2022 - Present**
Graduate Student Researcher, Teaching Assistant

- Segmenting hyperspectral observations of molecular and atomic gas into cloud catalogs in order to measure the rate of angular momentum transport between the warm and cold phases of the ISM
- Modeling the three-dimensional structure of molecular clouds to derive the effects of morphology on star-formation efficiency
- Teaching students in introductory and upper-division courses skills in programming, image processing, data analysis, and scientific presentation using custom-made worksheets and Jupyter notebooks

Flatiron Institute, Center for Computational Astrophysics: *New York, NY* **2025**
Research Analyst

- Built parallel-processing pipelines to reduce, analyze, and visualize the outputs from a suite of 3D radiation-hydrodynamic simulations covering a range of physical and numerical parameters
- Characterized the bulk properties of feedback-driven winds and calculated the rainback time of gas carried by outflows

Pomona College: *Claremont, CA* **2018 - 2022**
Research Assistant, Laboratory Coordinator & Peer Mentor

- Created an adaptive-time-stepping module in a computer program simulating the detection of gravitational waves by LISA; compiled documentation and wrote a project manual to train future research assistants
- Calibrated and installed an echelle spectrograph at Brackett Observatory
- Ran labs for an introductory physics course; tutored peers in physics, computer science, and philosophy of science

Jet Propulsion Laboratory: *Pasadena, CA (remote)* **2019 - 2020**
Remote Observer

- Planned observations and remotely operated the telescope at Table Mountain Observatory to refine the trajectories of asteroids and near-Earth objects

IAU Office of Astronomy for Development (OAD): *Cape Town, South Africa (remote)* **2020**
Intern

- Collected, cleaned, and labeled social-network data for the inaugural DARA Big Data Hackathon

SELECTED PROJECTS

Maize Storage as an American Option - *The Erdős Institute* **Fall 2026**

- Valued a farmer's decision of when to sell stored maize by modeling it as an American option, as the course project for Quantitative Methods in Finance

PUBLICATIONS

Mekuria, T. & Imara, N. – "Kinematics of HI Envelopes Associated with Molecular Clouds" – *AJ* 171, 200 (2026)

Mekuria, T. et al. – "Outflows are Driven by Supernovae in a z = 10 UV-bright Galaxy" – in preparation

Mekuria, T., Imara, N. & Forbes, J. – "The Global Structure of Molecular Clouds. II. Volume-Density Distributions" – in preparation